МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ РОССИЙСКОЙ ФЕДЕРАЦИИ

ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ АВТОНОМНОЕ
ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ
«Санкт-Петербургский национальный исследовательский
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ФАКУЛЬТЕТ СИСТЕМ УПРАВЛЕНИЯ И РОБОТОТЕХНИКИ

# ЛАБОРАТОРНАЯ РАБОТА №3

по дисциплине

«Операционная система Linux»

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# Выполнение лабораторной работы

Генерация варианта:

```
~/Desktop » python lab3.py                                                    c0ris@c0rislaptop
Neo:  Then why can't I see what happens to her?
The Oracle:  We can never see past the choices we don't understand.
Neo:  Are you saying I have to choose whether Trinity lives or dies?
The Oracle:  No. You've already made the choice, now you have to understand it.
Neo:  No, I can't do that. I won't.
The Oracle:  You have to.
Neo:  Why?
The Oracle:  Because you're The One.
Neo:  What if I can't? What happens if I fail?
The Oracle:  Then Zion will fall. Our time is up. Listen to me, Neo. You can save Zion if you reach
The Source, but to do that you will need the Keymaker.
Neo:  The Keymaker?
The Oracle:  Yes, he disappeared some time ago. We did not know what happened to him until now. He's
 being held prisoner by a very dangerous program, one of the oldest of us. He is called the Meroving
ian, and he will not let him go willingly.
Neo:  What does he want?
The Oracle:  What do all men with power want? More power.
The Oracle:  Be there, at that exact time, and you will have a chance.
```

Файлы lastname_out, lastname_err, полученные одной командой:

```
~/Desktop » python lab3.py > lastname_out 2> lastname_err                     c0ris@c0rislaptop

~/Desktop » cat lastname_out                                                  c0ris@c0rislaptop
Link:  Yes sir?
Morpheus:  Given your situation, I can't say I fully understand your reasons for volunteering to ope
rate onboard my ship. However, if you wish to continue to do so, I must ask you to do one thing.
Morpheus:  I hope so. Now re-patch the main AC to the hard drives and stand by to broadcast.
Link:  Yes, sir.
Trinity:  Still can't sleep? You wanna talk?
Neo:  They're just dreams.
Trinity:  She's gonna call. Don't worry.
Link:  There you are.

~/Desktop » cat lastname_err                                                  c0ris@c0rislaptop
Link:  I understand, sir, it's just that… I'm scoping some serious sentinel activity up here.
Morpheus:  Link.
Link:  What's that, sir?
Morpheus:  To trust me.
Link:  Yes, sir, I will, sir… I mean, I do, sir.
Trinity:  If you're afraid of something,…
Neo:  I just wish… I wish I knew what I'm supposed to do. That's all. I just wish I knew.
```

Файл lastname_final, полученный перенаправлением ошибок в stdout:

```
~/Desktop » python lab3.py > lastname_final 2>&1                              c0ris@c0rislaptop

~/Desktop » cat lastname_final                                                c0ris@c0rislaptop
Zion Controller:  [?]Control. Zion Control, stand by for Gate 3 lockdown.
Zion Virtual Control Operator:  The Nebuchadnezzar is down. Bay 7.
Zion Controller:  Understood.
Morpheus:  Captain Mifune.
Mifune:  Captain Morpheus.
Morpheus:  Are you here to escort me to the stockade, Captain?
Mifune:  I'm just here to keep the peace.
Lock's Lieutenant:  Commander Lock demands…
Mifune:  *coughs*
Lock's Lieutenant:  …requests your immediate counsel, sir.
Morpheus:  Link.
Link:  Sir?
Morpheus:  I want the ship ready to go as soon as humanly possible.
Link:  Understood, sir.
Neo:  What is it between them?
Trinity:  Morpheus and Lock? Niobe.
Neo:  Captain Niobe?
```

Подсчитанное количество слов:

```
~/Desktop » wc -w < lastname_out
81

~/Desktop » wc -w < lastname_err
62

~/Desktop » wc -w < lastname_final
97
```

Сгенерированные задания на пайплайн:

```
~/Desktop » python lab3_2.py
a) The character that says a phrase is "Merovingian"
        b) The character that is talked with/about is  "Keymaker"
                c) Should contain punctuation "`"
                        d) The subject that is talked about "very"
```

Задание а) - фразы, где говорит Merovingian:

```
~/Desktop » cat the-matrix-reloaded.csv | grep ^Merovingian
Merovingian," Aha, here he is at last. Neo, the One himself, right? And the legendary Morpheus. And Trinity of course, si belle qu'elle me fait souffrir. I have heard so much, you honour me. Please,
 sit, join us. This is my wife, Persephone. Something to eat? Drink? Hmm… of course, such things are contrivances like so much here. For the sake of appearances."
Merovingian," Yes, of course, who has time? Who has time? But then if we do not ever take time, how can we ever have time? Château Haut-Brion 1959, magnificent wine, I love French wine, like I love
the French language. I have sampled every language, French is my favourite - fantastic language, especially to curse with. Nom de Dieu de putain de bordel de merde de saloperies de connards d'enculé
s de ta mère. You see, it's like wiping your arse with silk, I love it."
Merovingian," Hmph… I am a trafficker of information, I know everything I can. The question is, do you know why you are here?"
Merovingian," Oh yes, it is true. The Keymaker, of course. But this is not a reason, this is not a `why.' The Keymaker himself, his very nature, is means, it is not an end, and so, to look for him i
s to be looking for a means to do… what?"
Merovingian," But do you? You think you do but you do not. You are here because you were sent here, you were told to come here and you obeyed. [Laughs] It is, of course, the way of all things. You s
ee, there is only one constant, one universal, it is the only real truth"
Merovingian," No. Wrong. Choice is an illusion, created between those with power, and those without. Look there, at that woman. My God, just look at her. Affecting everyone around her, so obvious, s
o bourgeois, so boring. But wait… Watch - you see, I have sent her dessert, a very special dessert. I wrote it myself. It starts so simply, each line of the program creating a new effect, just like
poetry. First, a rush… heat… her heart flutters. You can see it, Neo, yes? She does not understand why - is it the wine? No. What is it then, what is the reason? And soon it does not matter, soon th
e why and the reason are gone, and all that matters is the feeling itself. This is the nature of the universe. We struggle against it, we fight to deny it, but it is of course pretense, it is a lie.
 Beneath our poised appearance, the truth is we are completely out of control. Causality. There is no escape from it, we are forever slaves to it. Our only hope, our only peace is to understand it,
to understand the `why.' `Why' is what separates us from them, you from me. `Why' is the only real social power, without it you are powerless. And this is how you come to me, without `why,' without
power. Another link in the chain. But fear not, since I have seen how good you are at following orders, I will tell you what to do next. Run back, and give the fortune teller this message"
Merovingian," Oh yes, it is. The Keymaker is mine and I see no reason why I should give him up. No reason at all."
Merovingian," Please, ma chérie, I've told you, we are all victims of causality. I drink too much wine, I must take a piss. Cause and effect. Au revoir."
Merovingian," Oh God, my God, Persephone how could you do this, you betrayed me [speaking French]!"
Merovingian," Cause? There is no cause for this, what cause?"
Merovingian," Lipstick? Lipstick? What craziness you are talking about woman, there is no lipstick."
Merovingian," Ai-ai-ai-ai-ai-ai, woman, this is nothing, c'est rien, c'est rien du tout. It's a game, it is only a game."
Merovingian," All right. All right. Let us find out where this goes. You two, get the Keymaker."
Merovingian," Handle us? You'll handle us? You know, your predecessors had much more respect."
Merovingian," Okay, you have some skill. Kill him."
Merovingian," You see, he's just a man."
Merovingian," Damn it, woman, you will be the end of me. Mark my words, boy, and mark them well. I have survived your predecessors, and I will survive you!"
```

Задание б) - фразы а), где упоминается Keymaker:

```
~/Desktop » cat the-matrix-reloaded.csv | grep ^Merovingian | grep Keymaker
Merovingian," Oh yes, it is true. The Keymaker, of course. But this is not a reason, this is not a `
why.' The Keymaker himself, his very nature, is means, it is not an end, and so, to look for him is
to be looking for a means to do… what?"
Merovingian," Oh yes, it is. The Keymaker is mine and I see no reason why I should give him up. No r
eason at all."
Merovingian," All right. All right. Let us find out where this goes. You two, get the Keymaker."
```

Задание в) - фразы б), где есть “`”:

```
~/Desktop » cat the-matrix-reloaded.csv | grep ^Merovingian | grep Keymaker | grep '`'
Merovingian," Oh yes, it is true. The Keymaker, of course. But this is not a reason, this is not a `
why.' The Keymaker himself, his very nature, is means, it is not an end, and so, to look for him is
to be looking for a means to do… what?"
```

Задание г) - фразы в), где говорится “very”, их количество, найденное с помощью wc и grep:

```
~/Desktop » cat the-matrix-reloaded.csv | grep ^Merovingian | grep Keymaker | grep '`' | grep very
Merovingian," Oh yes, it is true. The Keymaker, of course. But this is not a reason, this is not a `
why.’ The Keymaker himself, his very nature, is means, it is not an end, and so, to look for him is
to be looking for a means to do… what?"

~/Desktop » cat the-matrix-reloaded.csv | grep ^Merovingian | grep Keymaker | grep '`' | grep very |
 wc -l
1

~/Desktop » cat the-matrix-reloaded.csv | grep ^Merovingian | grep Keymaker | grep '`' | grep -c ver
y
1
```

Все фразы для актёра Merovingian:

```
~/Desktop » cat the-matrix-reloaded.csv | grep ^Merovingian | sed "1i\--Text for $name" | head -5
--Text for Lambert Wilson
Merovingian," Aha, here he is at last. Neo, the One himself, right? And the legendary Morpheus. And
Trinity of course, si belle qu’elle me fait souffrir. I have heard so much, you honour me. Please, s
it, join us. This is my wife, Persephone. Something to eat? Drink? Hmm… of course, such things are c
ontrivances like so much here. For the sake of appearances."
Merovingian," Yes, of course, who has time? Who has time? But then if we do not ever take time, how
can we ever have time? Château Haut-Brion 1959, magnificent wine, I love French wine, like I love th
e French language. I have sampled every language, French is my favourite - fantastic language, espec
ially to curse with. Nom de Dieu de putain de bordel de merde de saloperies de connards d’enculés de
 ta mère. You see, it’s like wiping your arse with silk, I love it."
Merovingian," Hmph… I am a trafficker of information, I know everything I can. The question is, do y
ou know why you are here?"
Merovingian," Oh yes, it is true. The Keymaker, of course. But this is not a reason, this is not a `
why.’ The Keymaker himself, his very nature, is means, it is not an end, and so, to look for him is
to be looking for a means to do… what?"
```

Замена имени персонажа на имя актёра:

```
~/Desktop » counter=1; while read line; do echo $counter $line; ((counter++)); done < mer_text
1 Lambert Wilson:" Aha, here he is at last. Neo, the One himself, right? And the legendary Morpheus. And Trinity of course, si belle qu’elle me fait souffrir. I have heard so much, you honour me. Pl
ease, sit, join us. This is my wife, Persephone. Something to eat? Drink? Hmm… of course, such things are contrivances like so much here. For the sake of appearances."
2 Lambert Wilson:" Yes, of course, who has time? Who has time? But then if we do not ever take time, how can we ever have time? Château Haut-Brion 1959, magnificent wine, I love French wine, like I
love the French language. I have sampled every language, French is my favourite - fantastic language, especially to curse with. Nom de Dieu de putain de bordel de merde de saloperies de connards d’e
nculés de ta mère. You see, it’s like wiping your arse with silk, I love it."
3 Lambert Wilson:" Hmph… I am a trafficker of information, I know everything I can. The question is, do you know why you are here?"
4 Lambert Wilson:" Oh yes, it is true. The Keymaker, of course. But this is not a reason, this is not a `why.’ The Keymaker himself, his very nature, is means, it is not an end, and so, to look for
him is to be looking for a means to do… what?"
5 Lambert Wilson:" But do you? You think you do but you do not. You are here because you were sent here, you were told to come here and you obeyed. [Laughs] It is, of course, the way of all things.
You see, there is only one constant, one universal, it is the only real truth"
6 Lambert Wilson:" No. Wrong. Choice is an illusion, created between those with power, and those without. Look there, at that woman. My God, just look at her. Affecting everyone around her, so obvio
us, so bourgeois, so boring. But wait… Watch - you see, I have sent her dessert, a very special dessert. I wrote it myself. It starts so simply, each line of the program creating a new effect, just
like poetry. First, a rush… heat… her heart flutters. You can see it, Neo, yes? She does not understand why - is it the wine? No. What is it then, what is the reason? And soon it does not matter, so
on the why and the reason are gone, and all that matters is the feeling itself. This is the nature of the universe. We struggle against it, we fight to deny it, but it is of course pretense, it is a
 lie. Beneath our poised appearance, the truth is we are completely out of control. Causality. There is no escape from it, we are forever slaves to it. Our only hope, our only peace is to understand
 it, to understand the `why.’ `Why’ is what separates us from them, you from me. `Why’ is the only real social power, without it you are powerless. And this is how you come to me, without `why,’ wit
hout power. Another link in the chain. But fear not, since I have seen how good you are at following orders, I will tell you what to do next. Run back, and give the fortune teller this message"
7 Lambert Wilson:" Oh yes, it is. The Keymaker is mine and I see no reason why I should give him up. No reason at all."
8 Lambert Wilson:" Please, ma chérie, I’ve told you, we are all victims of causality. I drink too much wine, I must take a piss. Cause and effect. Au revoir."
9 Lambert Wilson:" Oh God, my God, Persephone how could you do this, you betrayed me [speaking French]!"
10 Lambert Wilson:" Cause? There is no cause for this, what cause?"
11 Lambert Wilson:" Lipstick? Lipstick? What craziness you are talking about woman, there is no lipstick."
12 Lambert Wilson:" Ai-ai-ai-ai-ai-ai, woman, this is nothing, c’est rien, c’est rien du tout. It’s a game, it is only a game."
13 Lambert Wilson:" All right. All right. Let us find out where this goes. You two, get the Keymaker."
14 Lambert Wilson:" Handle us? You’ll handle us? You know, your predecessors had much more respect."
15 Lambert Wilson:" Okay, you have some skill. Kill him."
16 Lambert Wilson:" You see, he’s just a man."
17 Lambert Wilson:" Damn it, woman, you will be the end of me. Mark my words, boy, and mark them well. I have survived your predecessors, and I will survive you!"
```

# Вывод по лабораторной работе

Благодаря лабораторной работе я научился работать с перенаправлениями и пайплайнами, задавать переменные окружения, ознакомился с командами grep, sed, wc. Сложные команды bash, на мой взгляд, писать довольно-таки быстро и просто, в любой момент ты выполнить достаточно непростую задачу всего лишь в пару команд, идущий друг за другом. Однако после их выполнения и получения нужных результатов ты вряд ли будешь их сохранять, передавать и многократно использовать в дальнейшем, ведь написанный код читается любым другим человеком с большими усилиями.